Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

MLOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, building SageMaker ML pipelines and the Terraform and CI/CD that run them. Graduated the internal MLOps SME Program by building a SageMaker pipeline for medical image classification, since published by AWS as an official sample: clinical quality and fairness gates, blue/green canary endpoints with auto-rollback, and PSI drift detection that starts retraining, all in Terraform + CodePipeline. Deep IaC and CI/CD foundation from DevOps engagements at RWS, State Street, and DTCC. Built LeetCode Rating Predictor and fixed a train/serve skew that had tripled its backtest error (MAE 38.76 to 12.46 on 57 contests). Open-source contributor to Feast, Apache Airflow, and Prefect. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems solved.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **MLOps Pipeline – SME Program Graduate (Jun 2025 – Aug 2025):** Architected end-to-end SageMaker Pipelines platform for medical image classification – training, model registry, blue-green canary deployment with automatic rollback on error and latency alarms, auto-scaling, and PSI drift detection. Provisioned full infrastructure as code via Terraform + CodePipeline; exposed inference through API Gateway + Lambda with a CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0; see Publications).
- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (∼90%) building the AWS foundation for a data-centre exit and a second landing zone in Terraform: 4,000+ resources across 20+ Terraform Cloud workspaces, none failing.
  - Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines.
  - Delivered two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect, plus self-service onboarding that gives each new workload its route table within 15 minutes.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **Internal Impact:** Published a second AWS sample on Organizations governance (see Publications). Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Delivered 2 org-wide tech talks on Terraform testing and GitOps pipeline gating. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for production observability.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS to embed continuous code quality gates into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Projects

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live

◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API GitHub

◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub

◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

Claude Cost Optimizer Mar 2026
Claude Code Plugin, MCP, TypeScript, React, Python GitHub | Live

◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

Other Notable Projects: itr-agent (deterministic tax engine, 151 tests), GroundZeroNN (neural network from scratch in NumPy), SelfHub (MCP personal-memory server), Instagram Autopilot (source-grounded animated SVG Reels on a GitHub Actions schedule; Bedrock writes the topic and caption).

Open Source Contributions (11 merged upstream)

- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

MLOps & ML Platform: SageMaker Pipelines, Model Registry, Endpoints (real-time & async), Custom PSI drift jobs, Blue-Green Deployment, A/B Traffic Shifting, Feature Stores (Feast)
ML Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Experiment Tracking, Hyperparameter Tuning
CI/CD for ML: CodePipeline, CodeBuild, GitHub Actions, Docker, Model Versioning, Automated Retraining Triggers
Cloud & Infrastructure: AWS (Lambda, API Gateway, ECS Fargate, CloudFront, EventBridge, S3, CloudWatch), Terraform, HCL, tftest
Data & Serving: FastAPI, GraphQL, REST, Data Pipelines, SHA-256 Deduplication, SQLite, X-Ray, CloudWatch Metrics
Languages: Python, HCL, SQL, TypeScript, JavaScript, Bash